

EFY Design Contest

Hello EFY! Let me start by saying this: "I am a big fan of EFY." I am an ECE, B.Tech 2014 pass out. In my college days I used to visit our library just to read EFY magazine.

You are the inspiration for doing 20+ projects in my academic years and to achieve merit prizes from NITs and big universities. You can see some of my projects at <http://www.waytoem-bedded.blogspot.in/>

I saw announcement about 'Electronics For You Design Contest' on your site some time ago. I want to exhibit two major projects for this competition: (1) Wireless Sensor Network Based Home Monitoring System and (2) Quad-copter for Live Streaming and Communication, which I hope to complete by November end.

M. Sundeeep Goud
Hyderabad

EFY: The contest was announced last year and the winners were awarded during Electronics For You Expo held in February this year. The next contest has not yet been announced. But you also have the option of submitting your projects, which seem to be very interesting, for publication in EFY magazine.

Embedded Projects

I am an engineering student and I need to design a mega project on embedded systems. Can you please help me with some of the latest designs and descriptions of the projects?

Sangpriya Tiwari
Through e-mail

EFY: You will find quite a few projects in previous issues of EFY magazine. Also, we give hands-on training on some subjects through EFY Tech Centers (see efytechcenter.com).

Temperature Controller for Small Cooling Fans

The circuit in August issue drives a cooling fan (12V, 1.2W) and the negative temperature resistors are placed on hot items. These reduce in resist-

'Spot An Error' Award Winners

In 'Perfume Vaporiser Adaptor' circuit published in August issue, C1 (150nF) is used in the circuit diagram and also mentioned in parts list but is missing in the component layout for the PCB. It should be placed between the LED and diac DB3.

Akkanagamma S.M.

□ In 'Perfume Vaporiser Adaptor' circuit published in August issue, it is written that, "It is an inexpensive solid-state heat controlling circuit built around the commonly available triac BT139 (TRIAC1)." It should be BT136.

Somnath Biswas

□ In the eStyle First Look section in August issue, under 'Videocoin Introduces VPhone,' the display is mentioned incorrectly as QVCA instead of QVGA display.

Siddharth Kaul

ance when the items get hot and the fan increases in speed. The fan takes 100mA when connected to the 12V supply. The whole circuit is over-designed and quite unnecessary.

First, what is the function of R8? The text says the voltage across this resistor will be up to 1.35V. But when 100mA flows, the maximum voltage will be 100mV. So why specify 1-watt resistor?

Second, what is the function of the 10-ohm resistors? The resistance (or impedance) of the fan is about 120 ohms. The NTC resistors will drop to 42 ohms when all are detecting 100°C. This makes the whole resistor network equal to about 10 ohms. If you remove the 10-ohm resistors, the NTC resistors only have to detect an additional 10°C to achieve the same overall resistance. Or you can use 330R NTC resistors.

Third, what is the function of diode D1; what is it protecting?

Fourth, what is the function of LED2? The fan will always be operating.

Colin Mitchell
Australia

The author, Petre Tzv Petrov, replies:

The circuit was meant to illustrate the basic principles and for some educational value. One can apply or not apply the methods and the design principles listed below at own risk:

1. The circuit illustrates the simplest thermal controller with NTC or

PTC and a DC motor.

2. We connect several NTCs (NTC1 to NTC4) in parallel to reduce the self-heating effect and to have good accuracy of the controller.

3. We use equalisation resistors (R2 to R5) to reduce the differences between the NTCs (if needed).

4. The usage of series resistor is for measurement of the current of the DC motor. The value of the resistor ($R8 = 1 \text{ ohm}$) is selected for fast and easy calculation of the current of the motor.

5. Usage of larger components is for easier and permanent attachment of the clips (connectors) of the measurement equipment. R8 is 1W resistor in that case. In case of blocking of the DC motor, or short circuit in the DC motor, that resistor will not burn immediately.

6. Usage of indicator LED2 is to show approximately the applied voltage over the DC motor and the speed of the DC motor.

7. Usage of resistor (R1) is to give some initial value of the current for the DC motor.

8. D1 protects the measurement and the on/off electronics for that circuit, and to some degree LED2.

9. The circuit is designed to resist blocking or short circuit of the DC motor, and some errors which could occur when it is implemented, especially when used for educational purpose or by non-experienced persons.